

Cocycle classification: cycle lattices, arithmetic torsion, and the tower wall

Draft (2026-08-10). Research report for the cocycle-classification track. This document makes no machine decision and does not change the observation language of `local-classes.md`.

Claim labels used throughout:

- **Verified** means checked against an accepted AIP, a mechanized repository fact, or the exact finite computation described in §2.4.
- **Desk-proved** means a complete mathematical argument is given here but is not mechanized.
- **Conjectured** means a precise plausible statement whose proof is not supplied.
- **Wall** means the requested conclusion is false as stated or the attempted method stops at a named missing lemma.

1. Verdict first

Desk-proved. At a fixed grade, formal-difference cocycle classes have a complete Smith-normal-form classification. For the reachable augmented configuration graph G , let

$$z : H_1(G; \mathbb{Z}) \rightarrow \mathbb{Z}^\Sigma$$

send a signed cycle to its signed letter-count vector. Then

$$\mathcal{Q}_G = H^1(G; \mathbb{Z}) / z^* \operatorname{Hom}(\mathbb{Z}^\Sigma, \mathbb{Z})$$

is the group of integral edge costs modulo potentials and static letter prices. If $K = \ker z$ and $B = \operatorname{im} z$, there is a short exact sequence

$$0 \rightarrow \operatorname{Ext}^1(\mathbb{Z}^\Sigma / B, \mathbb{Z}) \rightarrow \mathcal{Q}_G \rightarrow \operatorname{Hom}(K, \mathbb{Z}) \rightarrow 0.$$

The right side is the free **scheduling** part: it measures equal-count holonomy. The left side is the finite **arithmetic** part:

$$\operatorname{Ext}^1(\mathbb{Z}^\Sigma / B, \mathbb{Z}) \cong \operatorname{Hom}(\operatorname{Tor}(\mathbb{Z}^\Sigma / B), \mathbb{Q}/\mathbb{Z}).$$

Thus the scheduling/arithmetic split proposed in `local-classes.md` is not merely descriptive; it is the canonical fixed-grade exact sequence.

Verified, finite computation. Exact enumeration and SNF reproduce all four examples with no example-specific witness in the calculation:

- first touch has class order 1 at grades one and two;
- dirty write-back has infinite order at grades one and two;
- visit parity has exact order 2 at grades one and two;
- warmth has order 1 at grade one and infinite order from grade two.

The parity ambient quotient contains $(\mathbb{Z}/2)^2$ at both computed grades. Dirty and warmth land in the free scheduling quotient. First touch is a gradient.

Wall, requested axiom theorem; verified counterexample. Divisible coefficients, idempotent observer updates, and radius-zero events do **not** reduce local classes to homomorphisms plus write-back. The radius-zero two-state dirty observer itself has a grade-one local-table quotient of free rank three over $\mathbb{Q}$. Besides dirty up, charging read while dirty is independent: the two paths

$$\begin{aligned} A &= \text{down0readwrite1up}, \\ B &= \text{down0write1readup} \end{aligned}$$

have the same start, endpoint, letter counts, and write-back cost, but their dirty-read charges are 0 and 1. No coefficient divisibility can remove an equal-count holonomy difference. Allowing ParentFill adds warmth as another write-free free class rather than repairing the theorem.

Desk-proved, partial transgression theorem. For a genuine connected regular finite observer cover $Y \rightarrow X$ with deck group D , the usual five-term sequence is exact in this graph setting and identifies invariant upstairs classes modulo pullbacks of **all** base classes with $H^2(D; \mathbb{Z})$. For finite D this is $\text{Hom}(D, \mathbb{Q}/\mathbb{Z})$; the every- r -th-visit class on the one-node return loop is its generator for $D = \mathbb{Z}/r$. Equation (2) is the corresponding exact statement after retaining only global letter prices.

Wall, transgression conjecture as stated. The two quotients are not the same in general. The machine quotient removes only pullbacks from the letter rose, whereas the five-term quotient removes every class from the unobserved base graph. Moreover, idempotent observers are deterministic finite lifts but not graph coverings because their updates are not invertible. “Arithmetic classes are exactly transgressions of observer coverings” is therefore true for the cyclic covering slice and false without these two qualifications.

Wall, tower stabilization. The named examples stabilize by grade two, but the proposed fixed-gadget theorem is not proved and is false for the ambient cohomology groups: their free ranks already grow from 84 to 6828 for first touch. For the finite-dimensional local-table image, stabilization would require a grade-compatible finite boundary summary of integral holonomy relations. Leaf-only reads and writes, path-wide WriteBelow, and child-wide ParentFill prevent the obvious grade embedding from preserving either labels or tables. That missing boundary-summary lemma is the precise obstruction.

2. Fixed grade by Smith normal form

2.1 From the free action category to a graph group

Fix one finite observer A and grade n . Let $G = G_{A,n} = (V, E)$ be the directed multigraph of reachable augmented action configurations, retaining a separate edge for each runnable addressed generator. The path category of G is free on its edges. An integral cocycle on that category is consequently an arbitrary edge cochain

$$c \in C^1(G; \mathbb{Z}) = \mathbb{Z}^E.$$

Desk-proved. Quotienting edge cochains by potentials gives ordinary graph cohomology:

$$C^1(G; \mathbb{Z}) / \delta C^0(G; \mathbb{Z}) \cong H^1(G; \mathbb{Z}) \cong \text{Hom}(H_1(G; \mathbb{Z}), \mathbb{Z}).$$

This statement uses signed cycles in the underlying multigraph. It does not need strong connectivity and therefore includes irreversible observers such as first touch. Static prices form $\mathbb{Z}^\Sigma$; the labeling $\ell : E \rightarrow \Sigma$ pulls them back to edge cochains. The formal class group is the cokernel

$$\mathcal{Q}_G = \text{coker} \left(\mathbb{Z}^\Sigma \xrightarrow{\ell^*} H^1(G; \mathbb{Z}) \right).$$

This is the precise form of the SNF method requested by the track.

Warning, desk-proved. Equation (5) is the ambient edge-cost group, not the claim that every edge cochain is a local table. If $\rho : \mathbb{Z}^D \rightarrow \mathbb{Z}^E$ is the reachable table-to-edge map, the local-table group is its image in $\mathcal{Q}_G$. A named surcharge is one element of that image. Thus “first touch is zero” means that element is zero; it does not mean the very large ambient group (5) vanishes.

2.2 The canonical scheduling/arithmetic exact sequence

Let $H = H_1(G; \mathbb{Z})$, $K = \ker z$, and $B = \text{im } z$, where $z : H \rightarrow \mathbb{Z}^\Sigma$ is signed letter count.

Theorem, desk-proved. Sequence (1) is exact and splits noncanonically.

Proof. The sequence

$$0 \rightarrow K \rightarrow H \rightarrow B \rightarrow 0$$

has free quotient B , so restriction of homomorphisms gives

$$0 \rightarrow \text{Hom}(B, \mathbb{Z}) \rightarrow \text{Hom}(H, \mathbb{Z}) \rightarrow \text{Hom}(K, \mathbb{Z}) \rightarrow 0.$$

The image of a static price inside the left term is its restriction from $\mathbb{Z}^\Sigma$ to B . Applying $\text{Hom}(-, \mathbb{Z})$ to $0 \rightarrow B \rightarrow \mathbb{Z}^\Sigma \rightarrow \mathbb{Z}^\Sigma/B \rightarrow 0$ gives

$$\text{coker}(\text{Hom}(\mathbb{Z}^\Sigma, \mathbb{Z}) \rightarrow \text{Hom}(B, \mathbb{Z})) \cong \text{Ext}^1(\mathbb{Z}^\Sigma/B, \mathbb{Z}).$$

Quotient (6) by static prices to obtain (1). Since $\text{Hom}(K, \mathbb{Z})$ is free, the extension splits, although no choice-free splitting is implied. Finally, for a finitely generated abelian group C , $\text{Ext}^1(C, \mathbb{Z})$ is canonically dual to its torsion through $\mathbb{Q}/\mathbb{Z}$, giving (2). $\square$

Desk-proved. A class is scheduling-trivial exactly when its holonomy vanishes on K , equivalently when equal-count signed cycles have equal cost. Such a class induces an integral functional $B \rightarrow \mathbb{Z}$. Its arithmetic order is the least positive denominator needed to extend that functional to $\mathbb{Z}^\Sigma$. This recovers both obstructions in `local-classes.md` §3.3 and proves that every arithmetic obstruction is torsion.

2.3 The matrix

Choose an undirected spanning forest T of G . For an edge cochain x , integrate x from the root of each component along T to obtain a tree potential p_x . For every non-tree edge $e : u \rightarrow v$, define its fundamental cycle residual

$$h_x(e) = x(e) + p_x(u) - p_x(v).$$

There are

$$b_1 = |E| - |V| + \# \text{ components}$$

such edges. The residuals identify (4) with $\mathbb{Z}^{b_1}$. Put the six letter-indicator residuals in the columns of

$$L \in \mathbb{Z}^{b_1 \times |\Sigma|}.$$

Desk-proved. If the nonzero Smith diagonal of L is $d_1 \mid \cdots \mid d_s$, then

$$\mathcal{Q}_G \cong \mathbb{Z}^{b_1-s} \oplus \bigoplus_{d_i > 1} \mathbb{Z}/d_i.$$

For a named cost with residual column c , its order is the positive generator of the projection to k of

$$\ker_{\mathbb{Z}}[L \mid -c] = \{(\lambda, k) : L\lambda = kc\}.$$

If that projection is zero, the class has infinite order. Equations (8) and (9) are independent of the chosen forest because changing the forest changes the cycle basis unimodularly.

2.4 Exact computation

The computation used the single addressed head needed by the four examples. For each grade it did the following.

1. Enumerate every base memory and every legal initial head position with all observer marks at q_0 .
2. Close that set under all six runnable generators using the exact event order of `local-classes.md` §2.1. The charge reads the pre-update mark; writes update every non-root node on the focused path.
3. Retain parallel generator edges, build an undirected spanning forest, and form (7) for the six letter cochains and the named table.
4. Compute the Smith diagonal of L and of the seven-column constraint (9) over exact integers.

Verified, finite computation. Every enumerated graph below is weakly connected. Here s is the rank of L ; `torsion` lists the non-unit Smith entries of L ; `order` is the named class order.

observer	grade	$ V $	$ E $	b_1	s	free rank	Tor	order
first	1	40	128	89	5	84	none	1
first	2	3248	10080	6833	5	6828	none	1
dirty	1	20	72	53	5	48	none	∞
dirty	2	272	992	721	5	716	none	∞
par- ity	1	48	160	113	5	108	2, 2	2
par- ity	2	7168	24576	17409	5	17404	2, 2	2
warmth	1	40	128	89	5	84	none	1
warmth	2	3248	10080	6833	5	6828	none	∞

The rank is five rather than six because on every signed cycle the total number of downs equals the total number of ups. Equivalently,

$$down0 + down1 - up = \delta(\text{depth}).$$

Desk-checked. The computed orders agree with direct identities. First touch is the change in the number of seen nodes. Twice parity is the price one on each down plus the potential counting zero marks. Dirty and warmth have the equal-count holonomy paths already established in `local-classes.md`; equation (9) shows no nonzero multiple can become a letter price plus a potential. Warmth is trivial at grade one because a filled leaf has no children for `ParentFill` to invalidate; depth two is the first grade at which its witness exists.

2.5 The nonnegative refinement

SNF works in the group completion. It does not by itself say which formal classes or trivializations are realizable by nonnegative costs.

For a nonnegative table $t \in \mathbb{N}^D$, define its realizable formal class to be the image of $\rho(t)$ in $\mathcal{Q}_G$. The full positive image is the affine semigroup

$$\mathcal{P}_{A,n} = q\rho(\mathbb{N}^D) \subseteq \mathcal{Q}_G,$$

where $q : \mathbb{Z}^E \rightarrow \mathcal{Q}_G$ is the quotient. For one named nonnegative cost c , define the realizable-price relation

$$R_c = \{(k, \lambda) \in \mathbb{N} \times \mathbb{N}^\Sigma \mid \exists \Phi \in \mathbb{Z}^V, kc + \Phi \circ s = \ell^* \lambda + \Phi \circ t\}.$$

Desk-proved. Integer potentials in (12) may be shifted by a componentwise constant into $\mathbb{N}^V$, so (12) is exactly the repository's exchange discipline. In cycle coordinates it is

$$R_c = \{(k, \lambda) \in \mathbb{N}^{1+|\Sigma|} \mid L\lambda = kc\}.$$

It is therefore a homogeneous Presburger-definable affine semigroup. This is the separate positive layer that SNF intentionally forgets.

Verified for the named examples. Write a price vector in the order down0, down1, up, read, write0, write1.

- First touch, at every grade: $R_c = \{(k, 0) : k \in \mathbb{N}\}$.
- Parity, at every positive grade: k must be even. For $k = 2m$, all nonnegative prices are

$$\lambda = (m - u, m - u, u, 0, 0, 0), \quad 0 \leq u \leq m.$$

Equation (10) explains the freedom to move price from downs to ups.

- Dirty write-back, at grades one and two: $R_c = \{(0, 0)\}$.
- Warmth: the first-touch cone at grade one, and $R_c = \{(0, 0)\}$ at grade two.

The same descriptions for dirty and warmth hold at every later grade by the uniform witnesses in §5.1.

3. The axiom lattice

The proposed restrictions are:

1. coefficients in the divisible ordered monoid $\mathbb{Q}_{\geq 0}$;
2. each primitive update $F, W, L, P : Q \rightarrow Q$ is idempotent;
3. radius zero, meaning $P = \text{id}$.

“Idempotent” here means each named update squares to itself. Stronger notions such as a commuting semilattice action or a single dirty bit are different axioms and are not silently substituted.

3.1 What each restriction really removes

Desk-proved. Divisible coefficients kill the arithmetic term (2): tensoring (1) with $\mathbb{Q}$ kills all finite torsion. They do not kill the free scheduling term: the rationalization of $\text{Hom}(K, \mathbb{Z})$. Consequently dirty, dirty-read, redundant-write, and warmth holonomies all survive averaging.

Wall, idempotence is too weak. Idempotence excludes the literal update $q \mapsto q + 1$ on a counter, but it does not force primitive updates to commute and does not collapse their generated transformation semigroup to a set/reset bit. Even the standard dirty bit has more priced events than dirty up.

Desk-proved. Radius zero removes the warmth construction because no event at a parent can invalidate its children. It leaves parity, dirty write-back, dirty-read, and redundant-write tables untouched.

3.2 All three axioms, without ParentFill

Use $Q = \{C, D\}$ with

$$F = L = C, \quad W = D, \quad P = \text{id}.$$

All updates are idempotent and radius zero. At grade one, after removing a static baseline for each letter, the reachable state-sensitive table entries are represented by

$$u_D, \quad r_D, \quad z_D, \quad o_D,$$

meaning dirty up, dirty read, dirty write0, and dirty write1.

Verified, finite SNF; desk interpretation. Their only relation modulo letter prices and potentials is

$$u_D + z_D + o_D = \text{write0} + \text{write1}.$$

Indeed, within each fill/leave epoch, the first write makes the node dirty, every later write sees D , and the final dirty up occurs exactly when at least one write occurred. Hence the number of all writes is the number of dirty writes plus the dirty-up indicator. The local-table quotient is torsion-free of rank three, with basis, for example,

$$[u_D], [r_D], [z_D].$$

The dirty-read independence witness is (3). For redundant write0, compare

$$\text{down0write0write1write0} \quad \text{and} \quad \text{down0write1write0write0}.$$

They have the same endpoint and counts; both remain dirty and end with bit zero, but the dirty-write0 counts are one and two. Thus the rank-three result persists over $\mathbb{Q}_{\geq 0}$.

Wall, target theorem. The conjunction of all three proposed axioms has already failed before adding a second observer or ParentFill. A theorem with one generator would need an additional restriction on **which events may be priced**, such as “only up may inspect the dirty bit,” not merely a restriction on state updates.

3.3 All three axioms, with ParentFill

Desk-proved. The dirty rank-three slice embeds by synchronous product into the language with P . The two-state warmth observer also has idempotent updates: $F = H$, $P = C$, and $W = L = \text{id}$. Its depth-two witness is write-free, so it is independent of every dirty class detected only by writes or dirty state. Thus the divisible group completion has rank at least four: three dirty-table directions plus warmth.

This does both resource slices requested by the track. Choosing whether invalidation is priced selects between “at least rank three” and “at least rank four”; it cannot select the proposed rank-one headline.

3.4 Each single omission

Verified/desk-proved summary. Each omission leaves a strict obstruction; none repairs the failed target.

omitted restriction	remaining restrictions	exact obstruction
divisibility	idempotent, radius zero	order-2 arithmetic class at grade one
idempotence	divisible, radius zero	free reset-counter holonomy
radius zero	divisible, idempotent	free warmth holonomy

For the first row, take $Q = \{0, 1, 2, 3\}$, $q_0 = 0$, $W = P = \text{id}$, and

$$F = (0, 0, 2, 2), \quad L = (3, 1, 1, 3),$$

where a tuple lists the four function values. Both F and L are idempotent, while $L \circ F$ has the reachable two-cycle $3 \leftrightarrow 1$. Charge `down0` exactly when its pre-fill state is 1.

Verified, finite SNF. The grade-one graph has 84 vertices and 264 edges; (9) gives exact class order 2. Thus “primitive updates are idempotent” does not imply “the generated dynamics have no counters.”

For the second row, take $Q = \{0, 1\}$, let F toggle, let W reset to zero, and let $L = P = \text{id}$. Charge a down whose pre-state is 1. From an all-zero root configuration, compare

$$\begin{aligned} A &= \text{down0updown0write0write0}, \\ B &= \text{down0write0updown0write0}. \end{aligned}$$

They have identical endpoints and counts, but costs one and zero. This is free scheduling holonomy, so rational divisibility does not remove it. The third row is the existing warmth pair with costs three and five.

Wall, complete axiom-lattice classification. The tables above classify the proposed rank-one theorem negatively and give exact lower bounds for all four regimes. They do not present the entire class monoid of every finite idempotent transformation semigroup. That broader classification would first need a stronger algebraic definition of “set/reset automaton”; the literal idempotence axiom admits arbitrary noncommuting idempotents such as (17).

4. Arithmetic classes and transgression

4.1 The graph-cover five-term sequence

Let $p : Y \rightarrow X$ be a connected regular finite graph cover with deck group D . Put $\pi = \pi_1(X)$ and $N = \pi_1(Y)$, so $1 \rightarrow N \rightarrow \pi \rightarrow D \rightarrow 1$. Graph fundamental groups are free and have cohomological dimension one.

Theorem, desk-proved. With trivial integral coefficients, the five-term sequence reduces to

$$0 \rightarrow H^1(X; \mathbb{Z}) \xrightarrow{p^*} H^1(Y; \mathbb{Z})^D \xrightarrow{\text{tr}} H^2(D; \mathbb{Z}) \rightarrow 0.$$

Proof. The standard restriction/transgression sequence for $1 \rightarrow N \rightarrow \pi \rightarrow D \rightarrow 1$ begins with $H^1(D; \mathbb{Z}) = 0$ because D is finite, and ends here with $H^2(\pi; \mathbb{Z}) = 0$ because π is free. This gives

(19). From $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ and vanishing of positive-degree finite-group cohomology with rational coefficients,

$$H^2(D; \mathbb{Z}) \cong H^1(D; \mathbb{Q}/\mathbb{Z}) = \text{Hom}(D, \mathbb{Q}/\mathbb{Z}).$$

□

4.2 The visit counter is literally a transgression

Restrict to one child and let $a = \text{down0up}$ be its return loop. The modulo- r observer replaces that loop by its connected r -fold cyclic cover. Its closed upstairs loop is a^r .

Desk-proved. The every- r -th-visit cost sends a^r to one. It is invariant under the deck action. Extending it to the base would require assigning a the value $1/r$, so its transgression is the generator of

$$H^2(\mathbb{Z}/r; \mathbb{Z}) \cong \mathbb{Z}/r.$$

Multiplying by r makes the extension integral, exactly reproducing the exchange potential and static down price in `local-classes.md` §4.5. This is the promised five-term explanation of visit torsion.

4.3 What survives for the full action graph

The machine quotient (5) does not divide by $p^* H^1(X; \mathbb{Z})$; it divides only by static prices pulled back along the labeling map to the six-letter rose. These agree on the one-loop slice above but not on a general action graph, whose base H^1 is much larger.

Desk-proved replacement theorem. The arithmetic subgroup of the machine quotient is exactly (2). Put

$$D_{\text{count}} = \text{Tor}(\mathbb{Z}^\Sigma/B).$$

Then arithmetic classes are precisely the characters $D_{\text{count}} \rightarrow \mathbb{Q}/\mathbb{Z}$. This finite group is the abelian **count monodromy**: it measures which integral letter-count functionals exist on realized cycles but require denominators to extend to all formal letter counts.

This is transgression algebra, and it is computed by the same non-unit Smith entries as §2. It need not equal the deck group of the full observer cover.

4.4 Precise wall on the conjecture

Wall. A deterministic observer projection has unique forward path lifting, but it is a topological graph cover only when every event update is invertible on the reachable fiber. First touch, dirty, and warmth use resets and therefore fail the incoming-star condition. They have finite state, not finite monodromy in the covering-space sense.

Wall. Even for permutation observers, (19) quotients by all base classes, while (5) quotients by only six letter prices. The conjecture becomes correct in either of two qualified forms:

1. for a genuine finite regular observer cover, invariant classes modulo all base cohomology are exactly the transgressions (19); or

2. for any fixed-grade labeled graph, arithmetic classes modulo word prices are exactly the count-monodromy characters (2).

Identifying these two groups requires an extra hypothesis that base cohomology itself is generated by letter prices. The dyadic action graph does not satisfy that hypothesis: the large free ranks in §2.4 are direct finite counterexamples.

5. The tower attack

5.1 What stabilizes in the examples

Desk-proved. The four named costs have uniform tower verdicts after grade two.

- First touch is a potential at every grade.
- Visit modulo r has exact order r at every positive grade: the repeated left-child loop proves nontriviality, and the sum of node-counter potentials proves the r -fold relation.
- Dirty write-back is nontrivial at every positive grade. Let $D = \text{down}0^n$ and $U = \text{up}^n$. Compare

$$DU D \text{write}0 \text{write}0 \quad \text{with} \quad D \text{write}0 U D \text{write}0.$$

Both paths have the same endpoint, memory, dirty path, and letter counts. The first has zero dirty ups and the second has n .

- Warmth is trivial at grade one and nontrivial for every $n \geq 2$. The depth-two witness remains valid inside every deeper tree: both paths perform the same fills of the depth-two nodes, so any additional child resets below them also agree.

Thus the examples have threshold at most two. This is evidence for a local theorem, not a proof for arbitrary finite observers.

5.2 Equivariant reduction does not compute the full group

The complete tree has a large automorphism group, and the observer table is address-blind. It is tempting to quotient configurations by tree symmetry and compute one orbit gadget.

Wall, verified finite obstruction. Cohomology does not commute with this orbit quotient. Non-invariant cycles survive in $H_1(G_n; \mathbb{Z})$, and the ambient free rank grows sharply:

$$84 \rightarrow 6828$$

already between grades one and two for the first-touch graph. The parity graphs grow from free rank 108 to 17404. An orbit graph can compute the invariant summand, not the full SNF in (8), unless all nontrivial symmetry representations are carried as coefficients. That coefficient system grows with the grade, defeating the fixed-gadget claim.

5.3 Mayer–Vietoris and the missing boundary object

Cut the depth- $(n + 1)$ tree into its root and two depth- n subtrees. A Mayer–Vietoris recurrence would need each subtree to expose a finite boundary summary from which all global holonomy and price-extension relations can be reconstructed.

Wall. The obvious summary “root head position plus root mark” is not closed under the operations:

- reads and writes are legal only at the grade-dependent leaf boundary;
- `WriteBelow` changes every marked node on a root-to-leaf path, coupling the boundary to unboundedly many internal nodes;
- `ParentFill` changes two child marks across the proposed cut;
- an exchange potential may inspect the entire augmented configuration and can carry unbounded integral offsets even though the observer is finite.

Consequently the grade- n graph is not a label-and-cost-preserving subgraph of grade $n + 1$. Replacing an old leaf operation by “down, operate, up” changes both the letter-count vector and the observer events. There is no canonical map under which the relation lattices form an ascending or descending chain. Noetherian stabilization of sublattices of a fixed $\mathbb{Z}^D$ therefore cannot yet be invoked.

5.4 The exact theorem still needed

For the fixed reachable table domain D define the relation lattice

$$\mathcal{R}_n = \{(t, \lambda) \in \mathbb{Z}^D \oplus \mathbb{Z}^\Sigma \mid \rho_n(t) - \ell_n^* \lambda \in \delta \mathbb{Z}^{V_n}\}.$$

Its positive part is Presburger at each grade. Tower triviality of t asks for one $\lambda \in \mathbb{N}^\Sigma$ with $(t, \lambda) \in \mathcal{R}_n$ for every n .

Conjectured, finite-boundary stabilization. For each observer whose update monoid is a finite commuting band and whose priced events are radius zero, there is an effectively computable $N(A)$ and a finite boundary transfer object such that (21), including its positive price projection, is determined for every $n \geq N(A)$ by one fixed transfer relation.

The commuting-band and radius-zero hypotheses are deliberately stronger than the failed axiom lattice. They rule out (17) and the explicit cross-boundary reset. Leaf-only operations remain the hard case.

Wall, current track. No such transfer object or pumping lemma has been constructed. In particular, storing only observer states is insufficient; the summary must preserve an integral lattice of holonomies and its nonnegative price cone. This is the precise obstruction to the requested all-grade theorem. The fixed-grade SNF is complete, but a fixed finite gadget does not currently answer every grade.

6. Lean rung list

The statements below preserve the subtraction-free public vocabulary of `lean/Adic/Cocycle.lean`. Integer matrices may be used inside decidability proofs without changing executable cost statements.

1. **Finite observer.** Define `Observer` with finite `Mark`, `init`, `fill`, `writeBelow`, `leave`, and `parentFill`, plus an augmented `ActionConfig` carrying one mark per geometric node.

2. **Event semantics.** Define the pre-observation dispatcher for `AddressedOp` and prove that its base projection is `actionStep`.
3. **Local cocycle.** Define `localCost table word config` and prove `localCost_isCocycle` in the action-configuration analogue of `IsCocycle`.
4. **Letter prices.** Define `letterCost : (AddressedOp k → Nat) → ...` and prove its cocycle law. State table triviality only as `Exchange (localCost table) (letterCost prices) potential`.
5. **Edge/word equivalence.** Prove that one-step exchange on every reachable edge is equivalent to word exchange. The forward induction should reuse the shape of `exchange_append`.
6. **Natural symmetry at fixed grade.** On a finite reachable graph, prove that reversing an exchange uses $K - \text{potential}$, with K the finite maximum. This internalizes AIP-5’s fixed-grade group-completion argument.
7. **Cycle criterion.** Build an undirected spanning forest and prove that an integer edge cochain is a gradient iff every fundamental residual (7) vanishes. Keep this lemma separate from directed closed-walk statements.
8. **Smith interface.** Define the letter residual matrix L and prove the presentation (8), either using a library SNF or a verified certificate checker for externally produced unimodular matrices.
9. **Positive cone.** State and prove the equivalence between natural `Exchange` feasibility and the nonnegative Diophantine system (13), using componentwise shifts to naturalize integer potentials.
10. **Calibration examples.** Mechanize first touch with its seen-count potential; parity nontriviality from the two-return loop; and the doubled parity exchange identity. These exercise zero and torsion.
11. **Scheduling examples.** Mechanize the short uniform dirty pair (20), the dirty-read pair (3), and the warmth paths. These exercise equal-count holonomy without the large sparse/dense witness.
12. **Exact-sequence theorem.** After the graph layer is stable, formalize (1) for finite free abelian groups. The covering-space transgression should be a later track; it is not needed for the machine’s SNF classifier.

Verified vocabulary alignment. Rungs 3–5 state costs with `IsCocycle`, `Exchange`, and `exchange_append`; no theorem statement requires subtracting natural costs. This matches AIP-5 §1B and the current `lean/Adic/Cocycle.lean` API.

7. Walls and design consequences

Desk-proved. Fixed-grade classification is complete at the formal group level: compute the cycle lattice, apply SNF, and intersect with the Presburger-positive cone when realizable prices matter. Equation (1) is the quotable theorem: local cost has a free scheduling part and a finite arithmetic part.

Wall. “Divisible + idempotent + radius zero implies write-back only” is mathematically false. The minimal repair is not a proof trick; the observer language must constrain priced events or impose a substantially stronger algebra, such as a commuting set/reset band together with a declared pricing surface.

Wall. “Arithmetic equals transgression” needs a quotient named. It is true for invariant classes modulo all base cohomology and, separately, true as count-monodromy duality modulo word prices. Equating those versions is false on the dyadic action graphs.

Wall. The all-grade result remains open at a finite-boundary transfer lemma, not at SNF. The ambient groups demonstrably do not stabilize. A future tower track should target the local relation lattices (21) under a narrower commuting-band observer, not attempt to stabilize all of $H^1(G_n)$.

Repository anchors

- AIP-5 §§0–2 and §4: action-category cocycles, exchange-form natural discipline, dirty holonomy, and the local-class question.
- `aips/draft/local-classes.md`: the observer event language, fixed-grade edge feasibility, and the four example witnesses.
- `lean/Adic/Cocycle.lean`: `IsCocycle`, `Exchange`, `exchange_append`, and `freeUp_exchange`.
- `lean/Adic/Machine.lean`: `ActionConfig`, `AddressedOp`, `actionStep`, and the six raw local operations used by the finite enumeration.